

The Role of Blockchain in Public Administration's Standardisation and Digitalisation Challenges

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Abstract

The digital transformation of public administration has shifted from document digitisation to comprehensive process re-engineering, with blockchain and artificial intelligence (AI) emerging as central technologies. This paper examines how blockchain's core features—immutability, decentralisation, and cryptographic security—interact with AI's analytical capabilities to address persistent challenges in legislative document management and administrative standardisation. While blockchain ensures data integrity and traceability, its integration with AI enables advanced document classification and the automated execution of administrative procedures through smart contracts. The Italian regulatory framework, notably the *Decreto Semplificazioni* (Law No. 12/2019) and recent AI-related initiatives, provides partial legal recognition for these technologies, yet significant ambiguities remain concerning governance and liability. Implementation is further hindered by interoperability constraints and institutional resistance. Drawing on case studies such as Estonia's e-governance model and Italy's x-Leges system, the paper argues that effective innovation requires a fundamental redesign of administrative processes, embedding blockchain–AI solutions within frameworks that enhance transparency, efficiency, and constitutional safeguards.

Keywords: blockchain; artificial intelligence; public administration; regulatory framework; Akoma Ntoso; LegalTech

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1. Introduction

The digital transformation of the public sector represents one of the most significant paradigms shifts in governmental operations over the past two decades. This evolution has been characterised by a progressive transition from paper-based procedures to fully digital processes, fundamentally altering the relationship between citizens, businesses, and public institutions.¹ At the core of this transformation lies the dematerialisation process, definable as the process that allows the elimination of paper documentary supports, through the conversion of paper documents into electronic/digital documents, capable of maintaining legal and evidentiary value. The dematerialisation journey has progressed through several distinct phases. Initially, it involved the simple conversion of paper documents into digital formats, maintaining the same organisational structures and workflows. However, it has gradually evolved towards a more comprehensive digital transformation that encompasses the complete re-engineering of administrative processes. This shift has been driven by the need to address the inherent limitations of paper-based systems, which are usually costly, environmentally unsustainable, lack transparency, present difficulties in sharing and archiving, require significant retrieval time, and are prone to errors and loss.² In the European context, this transformation has been guided by a series of strategic initiatives and regulatory frameworks. The European Digital Single Gateway, established by Regulation (EU) 2018/1724,³ represents a significant step towards the digitalisation of public services at a transnational level, providing citizens and businesses with access to information, administrative procedures, and assistance services.⁴ Meanwhile, at the national level, countries have developed their own digital infrastructures and enabling platforms, such as digital identity systems, electronic payment platforms, and centralised population registries.

The Italian regulatory framework for dematerialisation finds its primary reference in the Digital Administration Code (Codice dell'Amministrazione Digitale – CAD) of 2005,⁵ which explicitly addresses the concept of dematerialisation in Article 42:

Public administrations evaluate in terms of cost-benefit ratio the recovery on digital media of documents and paper acts for which conservation is mandatory

¹ Chantal Bompreszi and Monica Palmirani, 'Blockchain technology for legislation: preliminary considerations' (2024) 1 *Comparazione e diritto civile*.

² Liliana Ávila, Leonor Teixeira and Pedro Almeida, 'Promotion of administrative modernization through processes dematerialization' in *Encyclopedia of Information Science and Technology* (3rd edn, IGI Global Scientific Publishing 2015).

³ Regulation (EU) 2018/1724 establishing a single digital gateway to provide access to information, to procedures and to assistance and problem-solving services and amending Regulation (EU) No 1024/2012, <https://eur-lex.europa.eu/eli/reg/2018/1724/oj/eng>.

⁴ Regulation (EU) 2018/1724 of the European Parliament and of the Council of 2 October 2018 establishing a single digital gateway [2018] OJ L 295/1, <https://eur-lex.europa.eu/eli/reg/2018/1724/oj/eng>.

⁵ Decreto Legislativo 7 marzo 2005, n. 82, Codice dell'amministrazione digitale, https://www.agid.gov.it/sites/default/files/repository_files/leggi_decreti_direttive/dl-7-marzo-2005-82_0.pdf.

or appropriate and provide for the preparation of consequent plans for replacing paper archives with digital archives.

This legislative foundation has been further strengthened by subsequent guidelines from the *Agenzia per l'Italia Digitale* (Agency for Digital Italy; AgID) on the formation, management, and preservation of digital documents.

As public administrations continue to advance along this digitalisation trajectory, emerging LegalTech solutions such as blockchain and AI present new opportunities⁶ to enhance transparency, security, and efficiency in administrative processes.⁷ These technologies offer innovative solutions to longstanding challenges in public governance, including document authenticity verification, secure identity management, and streamlined inter-institutional communication.⁸

While the global digital transformation continues to accelerate, blockchain technology has emerged as a transformative innovation with particular relevance for the public sector. At its core, blockchain represents a distributed database architecture where data are replicated across a decentralised network of nodes, functioning without centralised coordination through peer-to-peer consensus mechanisms.⁹ The fundamental characteristics that render this technology particularly valuable for governmental applications encompass its inherent immutability, achieved through sophisticated cryptographic hashing and block linking, alongside robust security protocols implemented via asymmetric cryptography.¹⁰ The potential advantages that blockchain could deliver in governmental applications are manifold: first of all, the establishment of trust in information and processes through decentralised, shared databases that are verified on a peer-to-peer basis; the creation of trusted audit trails that enable the monitoring of data entry, usage, and access whilst increasing transparency; the delicate balancing of data confidentiality with sharing requirements; and significant efficiency improvements and cost reductions through automation of verification procedures.¹¹ The transformative potential of blockchain is further amplified through the integration of smart contracts, deterministic computer programs that can execute automatically according to pre-specified functions, thus

⁶ Davide De Lungo and Giovanni Rizzoni (eds), *Le assemblee rappresentative nell'era dell'intelligenza artificiale. Profili costituzionali* (Giappichelli 2025).

⁷ Bompreszi and Palmirani (n 2).

⁸ Luca Cervone, Monica Palmirani and Fabio Vitali, 'The Intelligible Contract' (2020) *Proceedings of the 53rd Hawaii International Conference on System Sciences*, <https://doi.org/10.24251/HICSS.2020.220>.

⁹ Bompreszi and Palmirani (n 2).

¹⁰ Ashok Kumar Yadav and Ramendra Kumar Bajpai, 'KYC optimization using blockchain smart contract technology' (2020) 4(3) *International Journal of Innovative Research in Applied Sciences and Engineering* 669, 671.

¹¹ Sana Zeba, Preetam Suman and Kanishka Tyagi, 'Types of Blockchain' in Rajiv Pandey, Sam Goundar and Shahnaz Fatima (eds), *Distributed Computing to Blockchain* (Academic Press 2023) 55–68.

enabling the automation of complex administrative procedures whilst maintaining transparency and accountability.¹²

Governments worldwide have demonstrated substantial interest in blockchain applications, with more than 200 use cases identified in the public sector as of late 2020, spanning domains as diverse as digital currency, land registration, identity management, notarisation, supply chain traceability, healthcare, education, and legal entities management.¹³ Notable exemplars include Estonia's blockchain implementation (operational since 2012) which underpins the majority of government services, Japan's blockchain-based system for processing government tenders, the United Kingdom's exploration of blockchain for land registry digitalisation, and Dubai's comprehensive initiative to transform document management and service processes.¹⁴ Notwithstanding the considerable potential, widespread adoption of blockchain within the public sector faces several persistent challenges, including scalability limitations, interoperability constraints, and lingering legal uncertainties regarding data protection and regulatory compliance.¹⁵

This paper explores the relationship between blockchain and the public sector, along with its integration with AI, with particular emphasis on the legislative domain which represents the highest expression of public governance. The research focuses specifically on applications relevant to document authenticity, process transparency, and inter-institutional coordination, whilst excluding broader applications in public procurement and citizen services that fall outside the legislative domain. It is important to note that this paper does not present advanced prototypical systems, concentrating instead on the legal aspects and proposing novel solutions for the literature that are practical in nature, for example the integration between x-Leges and blockchain, without providing the technical architecture that would properly fall within the remit of computer scientists. Evaluating both technological possibilities and regulatory boundaries, this research contributes to the ongoing discourse on digital transformation of legislative processes whilst acknowledging the constitutional requirements for separation of powers that any technological solution must respect.

2. Methodology and Research Design

This research adopts a normative-analytical methodology that combines doctrinal legal analysis with conceptual exploration of emerging technologies in the legislative

¹² Roman Evgenievich Spiridonov, 'Restricted Move--a smart contract description language to create and control finance instruments on DFinance blockchain platform' (2021) 1864(1) Journal of Physics: Conference Series, <https://dx.doi.org/10.1088/1742-6596/1864/1/012109>.

¹³ Marek Skotnica and others, 'Process Digitalisation using Blockchain: EU Parliament Elections Case Study' (MODELSWARD 2021).

¹⁴ Mohammad Malkawi, Muneer Bani Yaseen and Doaa Habeebalah, 'Ethereum blockchain-based e-voting system for Jordan parliament elections' (2023) 17(2) Applied Mathematics and Information Sciences 233.

¹⁵ Primavera De Filippi, Morshed Mannan and Wessel Reijers, 'The a legality of blockchain technology' (2022) 41(3) Policy and Society 358.

domain. The methodological framework operates at the intersection of public law, constitutional theory, and legal informatics, employing a multi-dimensional approach to assess blockchain and AI applicability within legislative processes.

The research design considers three complementary analytical dimensions. First, a comparative institutional analysis examines existing implementations of blockchain and AI in public sector contexts, particularly focusing on the Estonian e-governance model and the Italian x-Leges system as concrete manifestations of digital transformation in legislative processes. Second, a regulatory framework analysis systematically evaluates the European and Italian legal landscape, including the AI Act (Regulation EU 2024/1689), the eIDAS Regulation, and the Italian CAD, to establish the juridical foundations and operational parameters for technological deployment. Third, a constitutional compatibility assessment examines how blockchain-AI integration can be reconciled with fundamental constitutional principles, particularly the separation of powers doctrine and parliamentary autonomy requirements.

The legal-technical evaluation framework applied throughout this paper assesses blockchain's suitability for legislative procedures according to four key criteria: (1) the constitutional conformity with institutional autonomy requirements; (2) technical feasibility within existing digital infrastructures; (3) regulatory compliance with data protection and administrative law; and (4) practical implementability considering organisational and cultural factors. This methodology deliberately focuses on legal and conceptual dimensions rather than developing advanced technical prototypes, proposing practical solutions grounded in legal informatics literature while acknowledging that detailed technical architectures fall within the domain of computer science.

3. Blockchain and AI in the Public Sector: Opportunities and Applications

The modern implementation of blockchain technology was introduced in 2008 by Satoshi Nakamoto with the Bitcoin protocol, building upon earlier cryptographic concepts such as Haber and Stornetta's work on secure timestamping (1991) and Back's Hashcash (1997). This innovation has subsequently been expanded and diversified as part of the broader distributed ledger technologies (DLT) ecosystem and represents an important technological innovation characterised by several fundamental elements that make it particularly valuable for the public sector.¹⁶ First of all, blockchain's decentralised nature allows data to be stored and validated across multiple nodes without requiring a central authority, which significantly reduces the risk of single points of failure and potential bottlenecks in governmental operations.¹⁷ Secondly, the immutability feature, achieved through cryptographic hashing mechanisms and the chronological linking of blocks, ensures that once information is recorded, it cannot be altered retroactively without leaving an evident trace, thus establishing a tamper-resistant audit trail that is particularly valuable for maintaining

¹⁶ Bomprezzi and Palmirani (n 2).

¹⁷ De Filippi, Mannan and Reijers (n 16) 360.

the integrity of public records.¹⁸ Thirdly, blockchain offers enhanced transparency, as all transactions recorded on the chain are visible to all participants (with appropriate access permissions), thereby fostering trust in governmental processes while simultaneously preserving data confidentiality through selective disclosure mechanisms.¹⁹ These characteristics collectively address longstanding challenges in public sector, including data silos, verification difficulties, and the inefficiencies associated with paper-based systems or centralised digital repositories, making blockchain an increasingly attractive solution for governments seeking to modernise their administrative infrastructure and enhance service delivery.²⁰

The implementation of blockchain technology in legislative domains also necessitates careful consideration of its constitutional implications, particularly regarding the separation of powers doctrine and institutional autonomy. First of all, any technological infrastructure must be designed in accordance with the constitutional requirement that legislative, executive, and judicial branches maintain appropriate independence whilst enabling necessary coordination. The Italian Constitutional Court's ruling No. 235/2015 explicitly reinforces this requirement, emphasising that the decentralised management of legislative information is essential for preserving institutional pluralism against the concentration of governmental authority.²¹

This important constitutional consideration can be effectively solved through a multi-level blockchain architecture that mirrors the constitutional structure itself, consisting of distinct yet interconnected layers that preserve institutional independence while facilitating secure information exchange.²² The first level would consist of private institutional blockchains for internal deliberative processes within each parliamentary chamber or committee; the second level would implement a permissioned blockchain for inter-institutional coordination; whilst the third level would provide a public-facing blockchain for transparency and citizen access. This layered approach creates an appropriate balance between confidentiality and transparency, ensuring that each institution retains sovereignty over its internal functions whilst participating in secure communication channels with other governmental bodies.

Secondly, the application of AI within this framework must adhere to constitutional principles of parliamentary autonomy and free mandate.²³ These principles require technological equity amongst parliamentary groups and accessibility for all elected

¹⁸ Spiridonov (n 13).

¹⁹ Tahir Alyas et al, 'Multi blockchain architecture for judicial case management using smart contracts' (2025) 15 Scientific Reports, <https://doi.org/10.1038/s41598-025-92842-8>.

²⁰ This applies particularly to administrative procedures as discussed also in Malkawi, Yaseen and Habeebalah (n 15).

²¹ Simon, *Philosophy of democratic government* (University of Chicago Press 1951); the importance of this principle is extensively discussed in the 'Report on AI in Parliamentary Context' (December 2024), Section 4.

²² Monica Palmirani et al, *Report on AI in Parliamentary Context1* (Final Draft, University of Bologna December 2024).

²³ Pierfrancesco Bresciani and Monica Palmirani, 'Constitutional Opportunities and Risks of AI in the law-making process' (2024) 2 *Federalismi.it* 1, 8–13.

representatives regardless of their technical expertise. This necessitates a hybrid approach combining blockchain's immutability with the interpretive flexibility of human-supervised AI systems ('hybrid AI'), thereby ensuring that technological innovation enhances rather than undermines the constitutional foundations of legislative processes.

The integration itself of AI within legislative contexts necessitates rigorous adherence to fundamental constitutional principles and rule of law requirements. As clearly expressed by Palmirani's comprehensive framework, AI applications in parliamentary settings potentially affect several constitutional fundamental principles meriting protection. First, the principles of state sovereignty and parliamentary autonomy demand that legislative bodies maintain unequivocal control over deployed AI technologies, preventing external entities from exerting undue influence over democratic processes. Concurrently, the separation of powers doctrine requires implementation of stringent safeguards when AI systems facilitate oversight functions. The constitutional principle of continuity of power presents another critical dimension, as technological dependence might disrupt parliamentary operations during system failures, a consideration that inherently limits the degree of automation permissible in legislative processes. Furthermore, the principle of free mandate necessitates equal technological access across all parliamentary factions, thereby preventing algorithmic advantages from superseding electoral mandates. The integrity of democratic processes constitutes the final constitutional imperative, requiring transparent and unbiased AI systems that neither distort public participation nor misrepresent parliamentary activities.

These constitutional requirements also compel adoption of a hybrid architectural approach that integrates symbolic AI (encompassing semantic knowledge modelling and legal reasoning) with traditional methods, establishing technical frameworks built upon decades of legal informatics research. Accordingly, parliaments should implement specific regulations, even within their internal rule of proceedings, to mitigate these constitutional risks, particularly for implementations involving predictive modelling or public engagement functionalities. As Bresciani and Palmirani (2024) observe, 'Allowing technology to shape the legal domain without a solid foundation in theory of law risks weakening the values that underpin democracy and the rule of law';²⁴ therefore, ensuring AI alignment with constitutional frameworks safeguards technological advancement whilst preserving the foundational pillars of legal systems.²⁵

The implementation of blockchain technology in public sector has transitioned from theoretical discourse to practical application through various pilot projects and case studies worldwide, providing empirical evidence of its transformative potential. The most comprehensive implementation can be observed in Estonia's e-governance model, which employs the Keyless Signature Infrastructure (KSI) Blockchain as the

²⁴ Palmirani et al (n 23) 4.

²⁵ Pierfrancesco Bresciani and Monica Palmirani, 'AI Act and National Parliaments: How the EU is Regulating Democracy in the Digital Era' (2025) *Ragion Pratica* 1, 249.

backbone of its digital ecosystem. Developed in 2007, this pioneering system does not store data directly but functions as an integrity verification mechanism through cryptographic hashing, protecting numerous critical governmental registries including health records, property registries, business registries, and the judicial system. Additionally, the Estonian electronic identity (eID) system, secured by blockchain technology, enables citizens and residents to access a wide array of governmental and private sector services, whilst the innovative e-Residency programme extends these benefits beyond national borders.²⁶ Furthermore, recent developments in Latin America provide another evidence of growing institutional interest in blockchain applications for public governance,²⁷ with Chile's experimentation in blockchain-based solutions for land registry modernisation and the Central Bank of Chile's launch of a Digital Peso pilot programme in April 2024 demonstrating practical implementation beyond the European context.²⁸ Again, and more directly relevant to legislative processes, the Philippines Senate Committee on Science and Technology conducted public hearings in October 2025 on Senate Bill No. 1330, commonly known as the Blockchain the Budget Bill,²⁹ which proposes utilising blockchain technology to enhance transparency and traceability in the national budget allocation process, representing one of the few concrete legislative initiatives specifically targeting parliamentary functions through distributed ledger technology.

In the Italian context, the x-Leges project represents a project of considerable interest in the digitalisation of the legislative process, implementing a peer-to-peer architecture for the electronic transmission of legislative documents between four key institutions: the Senate, Chamber of Deputies, Presidency of the Council, and Ministry of Justice.³⁰ Despite not explicitly employing blockchain technology, x-Leges could be considered a pioneering precursor to blockchain-based systems for governmental document management. Despite not integrating a real 'blockchain', the project incorporated several foundational principles that would later become defining characteristics of blockchain implementations: a distributed peer-to-peer architecture ensuring institutional autonomy; cryptographic security mechanisms through certified email (PEC); immutable document identification via Uniform Resource Names (URN); and a sophisticated tracking system for document versioning and provenance.³¹ This

²⁶ Bompreszi and Palmirani (n 2).

²⁷ Maximiliano Appendino et al, 'Crypto Assets and CBDCs in Latin America and the Caribbean: Opportunities and Risks' (2023) IMF Working Papers 2023/037, <https://doi.org/10.5089/9798400234804.001>.

²⁸ Sethaput, Vijak, and Supachate Innet, 'Blockchain application for central bank digital currencies (CBDC)' (2023) 26 Cluster Computing 4, 2183.

²⁹ Wahyu Aditia Rawul, Maura Fadzikra Zahwa, Ray Roshan, Mawar, 'Blockchain Technology Utilization in Improving Transparency of Regional Revenue Budget Allocation' (2025) 20 Spirit Publik, <https://jurnal.uns.ac.id/spirit-publik/article/download/98362/pdf>.

³⁰ Luca Santis et al, 'The x-Leges System: Peer-to-Peer for Legislative Document Exchange' in Proceedings of the 8th International Conference on Current Trends in Web Engineering (2006) 231–242 https://doi.org/10.1007/11823100_21.

³¹ Fabio Massimi, 'x-Leges: Stato dell'arte e prospettive future' (Agenzia per l'Italia Digitale Workshop standard normativi – x-Leges, 25 maggio 2015),

system effectively demonstrated how decentralised approaches could preserve and improve institutional independence while enabling secure and efficient document exchange, creating a conceptual foundation³² that blockchain technology could further enhance and expand.

Additional projects such as the Italian Blockchain Service Infrastructure (IBSI) demonstrate the country's commitment to experimenting with distributed ledger technologies, aligning with the European strategy and involving numerous public and private entities in developing a blockchain ecosystem for public services. Beyond these examples, blockchain applications in public sector extend to diverse domains including digital guarantees in public procurement as evidenced by recent Italian legislative decrees which mandate blockchain verification for *fidejussory* guarantees. Decentralised identity management through projects like the European Blockchain Services Infrastructure (EBSI), the European counterpart of the IBSI, demonstrates how blockchain implementations could be important for public services management.

In the context of public sector, the DPP³³ represents a crucial innovation developed by the EU as part of the Ecodesign for Sustainable Products Regulation. It is operational through blockchain technology, which enables comprehensive tracking and transparency of products throughout their lifecycle. This technology allows manufacturers, retailers, consumers, and regulatory bodies to access vital information about products, including their origin, composition, environmental impact, and end-of-life management instructions.

Considering the essential characteristics of the DPP, it is possible to imagine the development of a 'legislative digital passport' as a novel concept for enhancing legislative processes.

The parallels between these concepts find their roots from several fundamental attributes: first, the verifiable traceability across multiple stakeholders help maintaining data integrity; secondly, the chronological documentation of all modifications and interventions during the lifecycle; thirdly, the interoperable data exchange between disparate systems with varying access permissions; and fourthly, the authoritative certification of origin and authenticity. In the legislative context, these characteristics could be valuable for tracking the evolution of legal texts from initial drafting through committee review, amendments, final approval, and subsequent modifications, whilst maintaining an immutable record of institutional

https://www.agid.gov.it/sites/default/files/repository_files/documentazione/20150525_x-leges_agid_f_massimi_v_1_0.pdf.

³² Presidenza del Consiglio dei Ministri – Dipartimento per gli affari giuridici e legislativi, 'Relazione sui risultati raggiunti nell'attuazione dei progetti Normattiva e x-leges e sulle loro prospettive di sviluppo' (2022), <https://www.normattiva.it/>.

³³ European Union, 'EU's Digital Product Passport: Advancing transparency and sustainability', <https://data.europa.eu/en/news-events/news/eus-digital-product-passport-advancing-transparency-and-sustainability>.

contributions. Projects such as Trace4EU³⁴ are already exploring similar blockchain-based architectural frameworks for implementing product traceability scenarios, which could serve as technical reference models for legislative document tracking. The development of a legislative digital passport inspired by the DPP concept could leverage blockchain's traceability and immutability to ensure transparency and verifiability throughout the legislative process, enhancing citizen access to legislative information and contributing to democratic trust.

4. Integration of Blockchain and AI in the Public Sector

The integration of AI within the public sector represents a burgeoning phenomenon that has evolved significantly in recent years, with numerous governmental entities implementing AI-driven solutions across diverse functional domains. Within the Italian context, the incorporation of AI systems into administrative procedures has progressed from initial experimentation to more substantial implementation, transforming effectively the traditional processes from 'acts with mere electronic support' to 'acts with electronic elaboration' capable of generating autonomous decisions based on complex algorithms.³⁵ AgID has taken proactive measures³⁶ to regulate this integration, publishing comprehensive guidelines for the public sector that align with the risk assessment framework delineated in the EU AI Act, extending beyond cursory identification of potential issues. Internationally, distinct patterns of implementation have emerged: the United States has deployed AI at the federal level to detect counterfeit pharmaceutical products, facilitate access to federal student loan programmes, and expedite disability benefit applications, whilst the American Congress has initiated experimental projects employing AI for database searches, text drafting, meeting summarisation, and specialised policy analysis. The progressive incorporation of AI within administrative functions demonstrates the real advantages across multiple dimensions, including enhanced efficiency through automation of repetitive activities, finding patterns, customisation of public services based on analytical insights, heightened security through real-time data analysis, and optimised resource allocation through predictive modelling.

One of the domains in which this integration may be particularly useful is the legislative sphere, where challenges of traceability, authenticity, and accessibility of documents exist. The convergence between blockchain and AI presents transformative possibilities for enhancing the Italian legislative process, particularly through the augmentation of existing systems such as x-Leges.³⁷ This project, initiated under the auspices of the Italian Government's Normattiva programme, exemplifies an innovative architectural approach to legislative document management, employing

³⁴ TRACE4EU, <https://trace4eu.eu/>.

³⁵ Nathalie A Smuha, *Algorithmic Rule By Law: How Algorithmic Regulation in the Public Sector Erodes the Rule of Law* (Cambridge University Press 2024).

³⁶ Massimi (n 32).

³⁷ Luca De Santis et al, 'The x-Leges System: Peer-to-Peer for Legislative Document Exchange'. *Lecture Notes in Computer Science* (Springer Nature 2006) 231–242.

a sophisticated peer-to-peer architecture that connects four principal institutional nodes: Senate, Chamber of Deputies, Presidency of the Council, and Ministry of Justice.³⁸ x-Leges incorporates several foundational principles that closely align with blockchain's philosophical underpinnings: a decentralised peer-to-peer architecture that preserves institutional autonomy, cryptographic security mechanisms facilitated through PEC, immutable document identification via URN, and sophisticated tracking systems for document versioning and provenance.³⁹

The potential integration of blockchain technology could substantially enhance this system by providing an immutable ledger of legislative documents that guarantees authenticity and prevents unauthorised modifications, while AI components could augment these capabilities through automated classification, annotation, and analysis of legislative texts. Such integration would address current operational challenges, including the absence of a unified document repository, which could create inefficiencies in document exchange processes and necessitates manual reconciliation of documents across disparate platforms.⁴⁰ The implementation of smart contracts, particularly those developed using specialised languages such as MOVE,⁴¹ could automate critical procedural aspects of the legislative workflow, including validation, timestamps, and authorisation controls.⁴² Furthermore, this hybrid system could create an authenticated, traceable record of a legislative document's entire life cycle, from initial drafting through committee review, amendment processes, final approval, and subsequent modifications. From a practical implementation perspective, this would require a multi-level blockchain architecture⁴³ comprising private institutional blockchains for internal deliberative processes, an inter-institutional coordination blockchain for official document transmission, and a public legislative blockchain for transparency and citizen access, each layer maintaining appropriate balances between confidentiality and transparency whilst preserving the constitutional separation of powers.

x-Leges is a tangible exemplar of digitalisation within the Italian legislative domain. The system's current architecture, whilst innovative in its implementation of peer-to-peer document transmission and cryptographic verification mechanisms, represents only the initial stage in a potentially transformative evolutionary trajectory that could

³⁸ De Santis et al (n 31).

³⁹ Massimi (n 32). Presidenza del Consiglio dei Ministri – Dipartimento per gli affari giuridici e legislativi (n 33).

⁴⁰ De Santis et al (n 31).

⁴¹ Move is a statically typed, resource-oriented programming language invented by Facebook (Meta) for the Diem blockchain that enables secure development of smart contracts and custom transactions with built-in management of digital assets, differing from other blockchain languages like Solidity through its first-class resources with move semantics, formal verification capabilities, limited mutability, and parallel transaction support, thereby providing enhanced security, preventing double-spending, and maintaining strict ownership control while reducing gas costs; <https://move-language.github.io/move/>.

⁴² Spiridonov (n 13)

⁴³ Alys et al (n 20).

be substantially enhanced through the integration of blockchain and AI technologies.⁴⁴ The confluence of these complementary technological paradigms offers unprecedented opportunities for redefining administrative decision-making processes, particularly functional for the legislative sphere. The integration of blockchain technology into the x-Leges framework would fundamentally transform document authenticity verification, establishing an immutable chronological record of legislative amendments whilst simultaneously enabling comprehensive document traceability throughout the entire legislative life cycle.⁴⁵

This immutability characteristic addresses one of the most significant limitations of the current system: the absence of cryptographically guaranteed document integrity which necessitates reliance on traditional trust mechanisms between institutional actors.⁴⁶ The adoption of structured markup standards such as Akoma Ntoso would further enhance this framework by enabling semantic interoperability and facilitating AI-driven document analysis. Within this context, AI presents complementary capabilities⁴⁷ that further amplify blockchain's transformative potential. AI algorithms could be deployed to perform sophisticated content analysis of legislative texts structured according to standards such as Akoma Ntoso, automatically identifying semantic relationships between provisions, detecting potential inconsistencies or contradictions with existing legislation, and classifying documents according to subject matter, regulatory impact, or constitutional relevance.⁴⁸ Furthermore, machine learning techniques could facilitate predictive analysis of legislative trends, enabling more informed decision-making and resource allocation within the legislative process. The synergistic implementation of these technologies would lead to substantial efficiency gains, reducing manual document processing requirements whilst simultaneously enhancing the quality and consistency of legislative outputs through automated validation procedures.⁴⁹

From a transparency perspective, a blockchain-based x-Leges implementation would enable unprecedented levels of procedural visibility, allowing citizens and institutional stakeholders to verify the authenticity and chronology of legislative documents without requiring intermediary validation. However, it is imperative to acknowledge the implementation challenges associated with such a transformation, particularly those related to institutional governance structures, regulatory frameworks, technical standards harmonisation, and the ethical implications of algorithmic decision-making in legislative contexts.⁵⁰ The successful deployment of a blockchain-enhanced x-Leges system would necessitate careful consideration of these factors, alongside a

⁴⁴ Massimi (n 32).

⁴⁵ Alyas et al (n 20).

⁴⁶ De Santis et al (n 31).

⁴⁷ Salvatore Sapienza and Monica Palmirani, 'Algorithmic Knowability: A Unified Approach to Explanations in the AI Act', Third World Conference on Explainable Artificial Intelligence, xAI 2025, Istanbul, 9–11 July 2025, 185–209.

⁴⁸ Smuha (n 36).

⁴⁹ Presidenza del Consiglio dei Ministri – Dipartimento per gli affari giuridici e legislativi (n 33).

⁵⁰ De Filippi, Mannan and Reijers (n 16).

graduated implementation strategy that respects constitutional principles regarding institutional autonomy and the separation of powers.⁵¹

5. Relevant Regulation at European Level

The EU has embarked upon an ambitious regulatory agenda to govern emerging technologies, establishing a comprehensive framework that balances innovation with fundamental rights protection.

First of all, it should be emphasised that when examining blockchain and AI technologies, thorough analysis of the preliminary regulatory frameworks becomes imperative to establish their juridical foundation and operational parameters. These technologies necessitate careful evaluation of existing legislative interventions to determine their scope of application, contextual limitations, and potential implementation pathways within public sector operations and legislative processes. Furthermore, it is important to underline how these regulatory frameworks interact with established administrative procedures whilst maintaining constitutional principles and institutional autonomy requirements, particularly when deployed in contexts of significant public governance importance.

At the forefront of this endeavour stands the AI Act, which represents the first comprehensive legal framework for AI globally.⁵² This legislation adopts a risk-based approach, categorising AI systems into four distinct tiers (unacceptable, high, limited, and minimal risk) alongside a specific category for general-purpose AI systems.⁵³ In addition to this framework, the Digital Services Act (DSA) and Digital Markets Act (DMA) establish unified regulations applicable throughout the EU, with the dual objectives of creating a safer digital environment that protects users' fundamental rights and establishing equitable conditions to facilitate innovation and competitiveness.⁵⁴

It is also essential to consider the important Regulation (EU) No 910/2014, commonly referred to as 'eIDAS'. This Regulation establishes the fundamental framework for electronic identification and trust services across the internal market, providing legal certainty for electronic transactions. Of particular importance within this framework is Article 41, which stipulates that electronic temporal validations cannot be denied legal effect or admissibility in legal proceedings solely due to their electronic format or because they fail to meet the requirements for qualified electronic time validations. The revised eIDAS 2 Regulation further extends this recognition to electronic ledgers,

⁵¹ Monica Palmirani and Fabio Vitali, 'Akoma-Ntoso for Legal Documents' in *Legislative XML for the Semantic Web: Principles, Models, Standards for Document Management* (Springer Netherlands 2011).

⁵² Regulation (EU) 2024/1183 of the European Parliament and of the Council of 11 April 2024 amending Regulation (EU) No 910/2014 as regards establishing the European Digital Identity Framework [2024].

⁵³ Smuha (n 36).

⁵⁴ *ibid.*

enshrining in Article 45 that ‘an electronic ledger shall not be denied legal effect or admissibility as evidence in legal proceedings solely on the grounds that it is in electronic form’.⁵⁵

In the Italian juridical context, the *Decreto Semplificazioni*, subsequently converted into Law n. 12/2019, represents a pivotal milestone in the legal recognition of blockchain technology and smart contracts.⁵⁶ Article 8-ter of this decree provides the first legal definition of distributed ledger technologies as

technologies and protocols that use a shared, distributed, replicable, simultaneously accessible, architecturally decentralized registry on a cryptographic basis, such as to allow the registration, validation, updating and archiving of data both in clear text and further protected by cryptography verifiable by each participant, not alterable and not modifiable.

Crucially, the same article defines smart contracts as ‘computer programs that operate on distributed registry technologies and whose execution automatically binds two or more parties based on predefined effects’. The third paragraph of Article 8-ter establishes that memorising a document through distributed ledger technologies produces the legal effects of electronic temporal validation as defined in Article 41 of Regulation EU n. 910/2014, thus conferring legal validity to documents recorded on blockchain, equating them to documents with electronic timestamp.⁵⁷

This legislative framework is further complemented by Article 30 of the proposed Data Act, which establishes essential requirements for smart contracts regarding robustness, access control, safe termination, data archiving, and functional consistency. The Italian regulatory landscape has also evolved in parallel for AI, with the recent adoption of *Disegno di Legge* n.1146 on AI, approved by the Italian Senate in March 2025. This legislation represents the first legislative proposal in Europe designed to implement the provisions of the European AI Act, introducing regulatory criteria and both general and sector-specific norms. Notably, the Italian approach exhibits a cautious and conservative orientation, emphasising the primacy of human intellectual labour and restricting AI system utilisation to supportive and instrumental activities.

The Italian regulatory orientation has been further reinforced by Law n. 132/2025, whose Article 14⁵⁸ explicitly codifies fundamental principles governing AI deployment within public sector contexts, mandating that AI systems operate exclusively as instrumental support to administrative decision-making whilst ensuring transparency through the traceability and intelligibility of AI functioning for affected parties, promoting a ‘human-on-the-loop’ paradigm wherein human autonomy and decisional authority remain paramount, and unequivocally attributing legal responsibility to the

⁵⁵ Regulation (EU) 2024/1183 (n 1).

⁵⁶ De Filippi, Mannan and Reijers (n 16).

⁵⁷ Bompreszi and Palmirani (n 2).

⁵⁸ Legge 23 settembre 2025, n. 132, art. 14.

public administration itself rather than to the technological systems it employs. The AgID has further contributed to this regulatory landscape by developing guidance on AI implementation for Italian public authorities, aligned with the risk assessment provisions set out in the AI Act.⁵⁹

Similarly, it is worth highlighting the Italian Chamber of Deputies expression of interest in the GENAI4LEX-B project, winning projects from a call dedicated to the use of AI in legislative processes. The project combines generative AI techniques with symbolic approaches to support regulatory research activities, drafting simplification, and legislative compliance verification.⁶⁰

6. Challenges and Implementation Barriers of Blockchain–AI Integration

The convergence of blockchain technology and AI within public sector contexts presents a complex landscape of transformative opportunities alongside substantial implementation challenges. Technical interoperability emerges as a critical concern, particularly as blockchain and AI systems must interface not only with each other but also with legacy administrative infrastructures.⁶¹ The heterogeneous nature of blockchain implementations, with varying protocols, consensus mechanisms, and architectural approaches, complicates the creation of unified systems where AI can effectively analyse cross-platform data. This integration challenge is exemplified by projects like x-Leges, where the potential enhancement of its peer-to-peer architecture with AI-powered analytical capabilities requires sophisticated technical solutions to preserve both immutability and analytical flexibility.⁶²

The regulatory landscape surrounding these converging technologies further shows implementation difficulties. While significant progress has been made through the EU AI Act and the blockchain provisions in the Italian *Decreto Semplificazioni*, considerable ambiguities remain regarding liability allocation in hybrid systems where AI algorithms interact with smart contracts to *partially* automate administrative decisions.⁶³ The question of how to attribute responsibility when an AI system incorrectly classifies legislative documents within a blockchain-based repository,

⁵⁹ Agenzia per l'Italia Digitale, Linee guida per l'adozione, l'acquisto, lo sviluppo di sistemi di intelligenza artificiale nella Pubblica Amministrazione (AgID 2025),

https://www.agid.gov.it/sites/agid/files/2025-02/Linee_Guida_adozione_IA_nella_PA.pdf.

⁶⁰ Monica Palmirani et al, 'Legal Drafting supported by AI: enhancing LEOS', Ital-IA 2024: 4th National Conference on Artificial Intelligence, Naples, May 2024, <https://ceur-ws.org/Vol-3762/590.pdf>; Monica Palmirani et al, 'GENAI4LEX-B: AI-powered legislative support in the Italian Chamber of Deputies', Interoperable Europe Platform - Public Sector Tech Watch, European Commission, Case Study PSTW-2194 (2024), <https://interoperable-europe.ec.europa.eu/collection/public-sector-tech-watch/genai4lex-b-ai-powered-legislative-support-italian-chamber-deputies>.

⁶¹ De Filippi, Mannan and Reijers (n 16).

⁶² Massimi (n 32).

⁶³ Regulation (EU) 2024/1183 of the European Parliament and of the Council of 11 April 2024 amending Regulation (EU) No 910/2014 as regards establishing the European Digital Identity Framework.

potentially triggering inappropriate smart contract executions, exemplifies the novel legal challenges that arise at this technological intersection.⁶⁴

Cultural resistance to organisational change presents perhaps a more formidable barrier than technical or regulatory challenges. The integration of these disruptive technologies requires fundamental reconceptualisation of administrative procedures, moving beyond mere digitalisation towards genuinely transformed processes that leverage AI's analytical capabilities within blockchain's secure and transparent framework.

This transformative potential directly confronts the inherent conservatism of many public institutions, where established hierarchies and workflows may be perceived as threatened by the decentralised paradigm of blockchain and the autonomous capabilities of AI systems.⁶⁵ A fundamental tension exists between the traditionally centralised nature of public administration and the decentralising architecture of blockchain, which is further complicated when AI systems introduce automated decision-making capabilities.⁶⁶ This disjuncture manifests in practical challenges regarding governance structures, where questions arise about how to maintain appropriate administrative controls while simultaneously embracing the efficiency gains offered by these technologies without removing the control power from humans. The multi-level blockchain architecture⁶⁷ proposed for legislative applications offers a potential resolution, creating a graduated system where AI systems can operate with varying degrees of autonomy across different administrative layers, dependent upon the sensitivity and significance of the processes being managed.

7. Conclusions, Key Findings and Future Research Directions

Despite the challenges identified above, the synergistic potential of blockchain and AI integration within public sector remains substantial. As demonstrated throughout this paper, AI's capability to analyse vast quantities of administrative data can be powerfully enhanced by blockchain's ability to ensure the provenance and integrity of that data. Similarly, blockchain's capacity to execute smart contracts can be significantly augmented by AI systems capable of adapting contract parameters based on emerging conditions and regulatory requirements.

The Estonian e-governance model demonstrates the practical benefits of this technological convergence, using blockchain to secure governmental registries whilst incorporating increasingly sophisticated analytical capabilities to enhance service delivery and regulatory compliance. The x-Leges system in Italy, while not explicitly employing blockchain technology, has created a foundation for peer-to-peer document exchange that could be substantially enhanced through blockchain

⁶⁴ Smuha (n 36).

⁶⁵ Cervone, Palmirani and Vitali (n 9).

⁶⁶ Spiridonov (n 13).

⁶⁷ Alyas et al (n 20).

integration, potentially transforming legislative document management through immutable, traceable records from drafting to implementation.

Successfully navigating this complex terrain requires careful balance between technological innovation and regulatory compliance, a balance that becomes particularly delicate when AI and blockchain technologies intersect. As AI systems evolve to interpret and classify legislative documents stored on blockchain platforms, questions arise regarding the appropriate degree of automation in administrative decision-making and the necessary human oversight mechanisms. These considerations are not merely technical but fundamentally involve questions of administrative values, constitutional principles, and democratic accountability.

As stated before, the transformative potential of these integrated technologies ultimately necessitates comprehensive reconceptualisation of administrative processes in digital terms. Rather than grafting emerging technologies onto existing procedures, truly transformative implementations require fundamental re-engineering that leverages AI's analytical capabilities within blockchain's secure and transparent framework. This approach recognises that the greatest value emerges not from the technologies individually, but from their thoughtful integration within redesigned administrative systems that maintain core governmental and constitutional values whilst enhancing efficiency, transparency and citizen engagement.

Future research in this domain could explore innovative governance models for hybrid systems where blockchain provides secure infrastructure for increasingly autonomous AI operations. Further investigation is needed into ethical frameworks for algorithmic decision-making within immutable blockchain contexts, particularly as they relate to legislative processes where constitutional principles must be preserved. Additionally, methodologies for evaluating the real-world impact of these integrated technologies on administrative efficiency, transparency and democratic accountability should be developed. The integration of both blockchain and AI in the public sector represents a promising area for further exploration, potentially offering a mechanism to track legislative documents throughout their lifecycle while ensuring verifiability and transparency.

Finally, as the regulatory landscape continues to evolve, ongoing research should address the interplay between technological capabilities and legal frameworks, ensuring that governance structures adapt appropriately to these transformative technologies while preserving fundamental constitutional values.

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